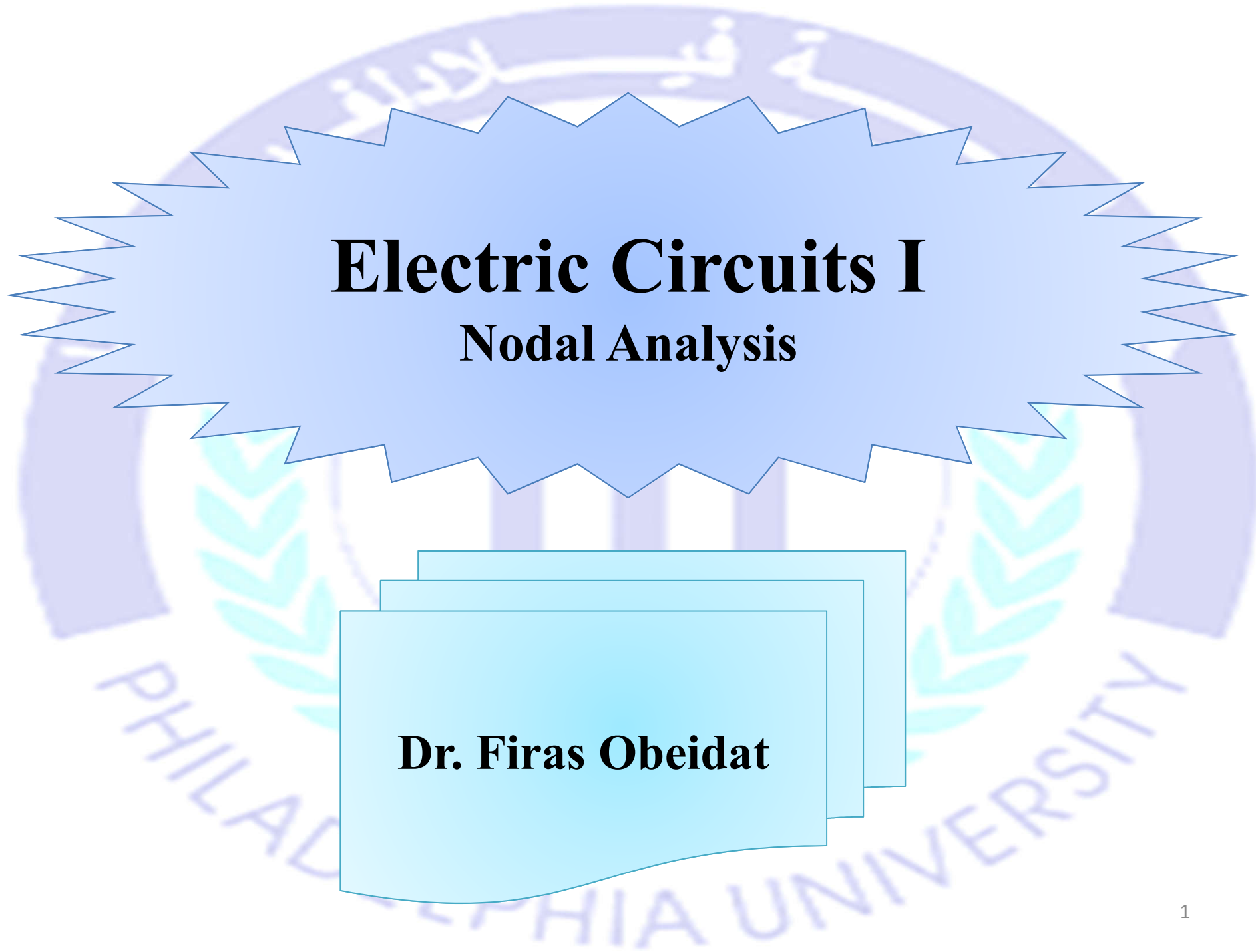




Electric Circuits I

Nodal Analysis



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Nodal Analysis Without Voltage Source

- **Nodal analysis, which is based on a systematic application of Kirchhoff's current law (KCL).**
- **A node is defined as a junction of two or more branches.**
- **define one node of any network as a reference (that is, a point of zero potential or ground), the remaining nodes of the network will all have a fixed potential relative to this reference.**
- **For a network of N nodes, therefore, there will exist $(N-1)$ nodes with a fixed potential relative to the assigned reference node.**

Nodal Analysis Without Voltage Source

Steps to Determine Node Voltages

- 1- Determine the number of nodes within the network.**
- 2- Select a node as the reference node. Assign voltages $V_1, V_2, V_3, \dots, V_{n-1}$ to the remaining nodes. The voltages are referenced with respect to the reference node.**
- 3- Apply KCL to each of the non-reference nodes. Use Ohm's law to express the branch currents in terms of node voltages.**
 - Assume that all unknown currents leave the node for each application of Kirchhoff's current law. In other words, for each node, don't be influenced by the direction that an unknown current for another node may have had. Each node is to be treated as a separate entity, independent of the application of Kirchhoff's current law to the other nodes.
 - Current flows from a higher potential to a lower potential in a resistor.

Nodal Analysis Without Voltage Source

Steps to Determine Node Voltages

This principle can be expressed as

$$i = \frac{v_{higher} - v_{lower}}{R}$$

4- Solve the resulting simultaneous equations to obtain the unknown node voltages.

Example: Calculate the node voltages in the circuit?

Node 1

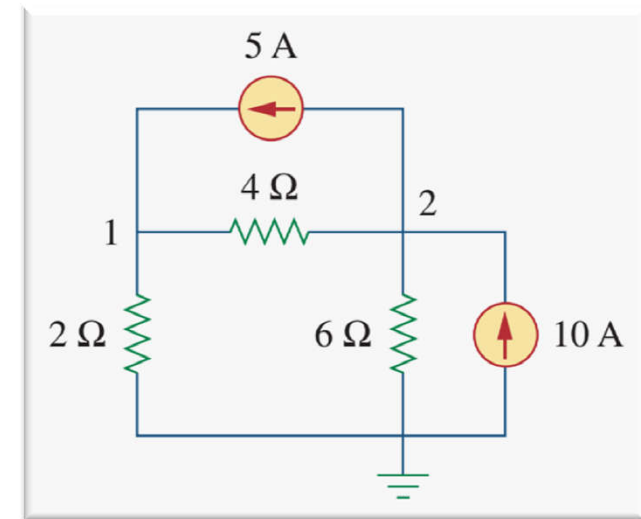
$$-5 + \frac{v_1 - v_2}{4} + \frac{v_1 - 0}{2} = 0 \quad (1)$$

Multiplying each term by 4, we obtain

$$3v_1 - v_2 = 20 \quad (2)$$

Node 2

$$5 + \frac{v_2 - v_1}{4} + \frac{v_2 - 0}{6} - 10 = 0 \quad (3)$$



Nodal Analysis Without Voltage Source

Multiplying each term by 12, we obtain

$$-3v_1 + 5v_2 = 60 \quad (4)$$

Using the elimination technique to solve equations (2) and (4) to get v_1 and v_2

$$3v_1 - v_2 = 20 \quad (2)$$

$$-3v_1 + 5v_2 = 60 \quad (4)$$

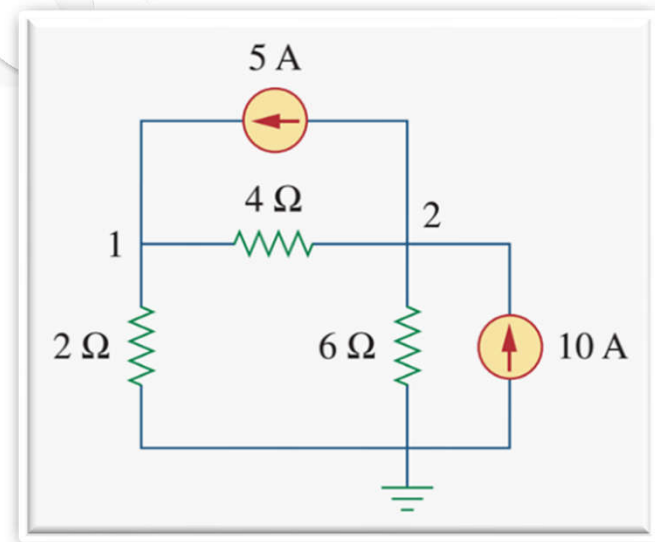
$$4v_2 = 80$$

$$v_2 = 20V$$

Substituting v_2 in equation (2)

$$3v_1 - 20 = 20 \quad (2)$$

$$v_1 = \frac{40}{3} = 13.33V$$



Nodal Analysis Without Voltage Source

Example: Calculate the node voltages in the circuit?

Node 1

$$-3 + \frac{v_1 - v_2}{2} + \frac{v_1 - v_3}{4} = 0 \quad (1)$$

Multiplying by 4 and rearranging terms

$$3v_1 - 2v_2 - v_3 = 12 \quad (2)$$

Node 2

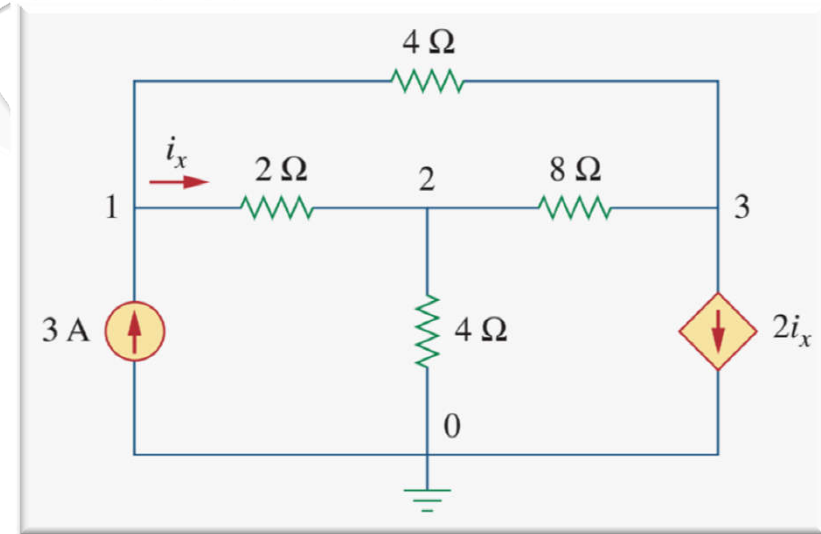
$$\frac{v_2}{4} + \frac{v_2 - v_1}{2} + \frac{v_2 - v_3}{8} = 0 \quad (3)$$

Multiplying by 8 and rearranging terms

$$-4v_1 + 7v_2 - v_3 = 0 \quad (4)$$

Node 3

$$2i_x + \frac{v_3 - v_1}{4} + \frac{v_3 - v_2}{8} = 0 \quad (5)$$



Nodal Analysis Without Voltage Source

$$i_x = \frac{v_1 - v_2}{2} \quad (6)$$

Substitute eq.(6) in eq.(5)

$$2\left(\frac{v_1 - v_2}{2}\right) + \frac{v_3 - v_1}{4} + \frac{v_3 - v_2}{8} = 0 \quad (7)$$

Multiplying by 8 and rearranging terms

$$6v_1 - 9v_2 + 3v_3 = 0$$

$$2v_1 - 3v_2 + v_3 = 0 \quad (8)$$

Solve equations (2), (4), and (8) using elimination technique to get v_1 , v_2 and v_3

Add eq.(2) to eq.(8)

$$3v_1 - 2v_2 - v_3 = 12 \quad (2)$$

$$2v_1 - 3v_2 + v_3 = 0 \quad (8)$$

$$5v_1 - 5v_2 = 12$$

$$v_1 - v_2 = \frac{12}{5} = 2.4 \quad (9)$$

Add eq.(4) to eq.(8)

$$-4v_1 + 7v_2 - v_3 = 0 \quad (4)$$

$$2v_1 - 3v_2 + v_3 = 0 \quad (8)$$

$$-2v_1 + 4v_2 = 0$$

$$v_1 = 2v_2 \quad (10)$$

Substitute eq.(10) in eq.(9)

$$2v_2 - v_2 = 2.4$$

$$v_2 = 2.4V \quad (11)$$

$$v_1 = 2v_2 = 2 \times 2.4 = 4.8V \quad (12)$$

Substitute eq.(11) in eq.(12) in eq.(8)

$$2 \times 4.8 - 3 \times 2.4 + v_3 = 0$$

$$v_3 = -2.4V \quad (8)$$

Nodal Analysis Without Voltage Source

Example: Calculate the node voltages in the circuit?

Node 1

$$-5 + \frac{v_1}{0.5} + \frac{v_1 - v_2}{0.4} + \frac{v_1 - v_3}{0.2} = 0 \quad (1)$$

$$9.5v_1 - 2.5v_2 - 5v_3 = 5 \quad (2)$$

Node 2

$$5 + \frac{v_2}{0.5} + \frac{v_2 - v_1}{0.4} + \frac{v_2 - v_3}{0.1} = 0 \quad (3)$$

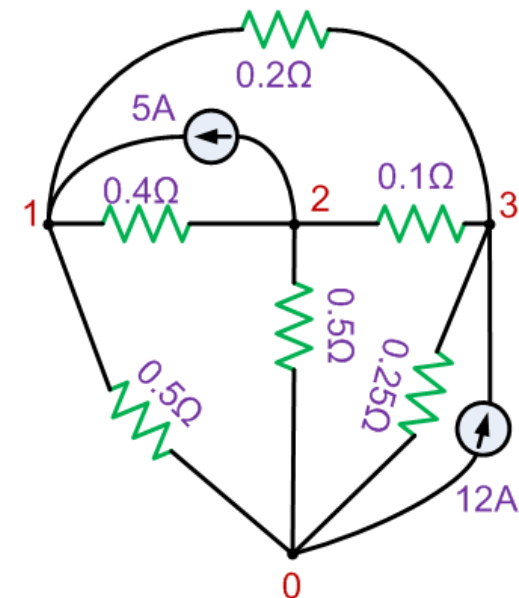
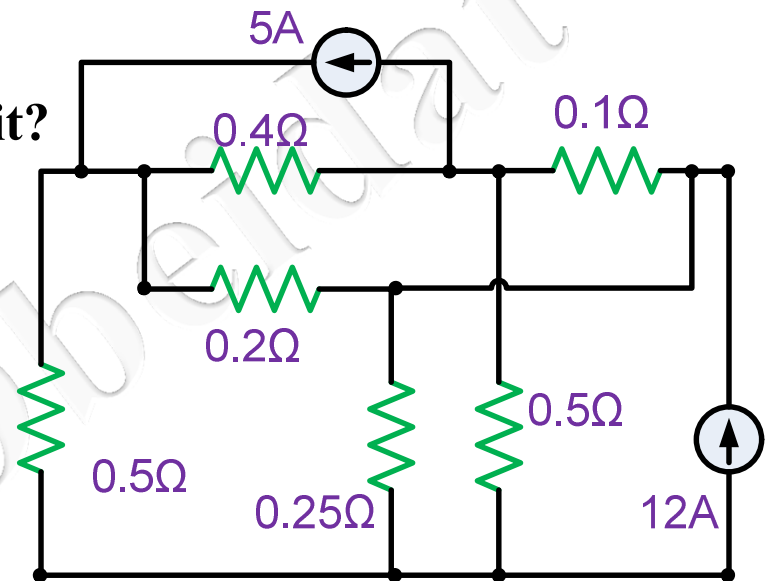
$$-2.5v_1 + 14.5v_2 - 10v_3 = -5 \quad (4)$$

Node 3

$$-12 + \frac{v_3}{0.25} + \frac{v_3 - v_1}{0.2} + \frac{v_3 - v_2}{0.1} = 0 \quad (5)$$

$$-5v_1 - 10v_2 + 19v_3 = 12 \quad (6)$$

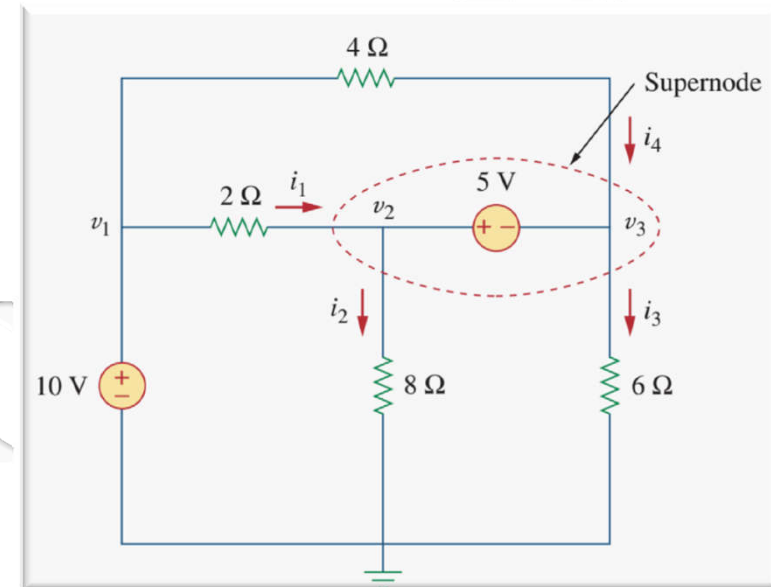
Solve eq.(2), eq.(4) and eq.(6) to get v_1 , v_2 , and v_3 .



Nodal Analysis With Voltage Source

CASE 1: If a voltage source is connected between the reference node and a non-reference node, we simply set the voltage at the non-reference node equal to the voltage of the voltage source. In the figure for example, $v_1 = 10V$.

CASE 2: If the voltage source (dependent or independent) is connected between two non-reference nodes, the two non-reference nodes form a generalized node or supernode; we apply both KCL and KVL to determine the node voltages.



A **supernode** is formed by enclosing a (dependent or independent) voltage source connected between two non-reference nodes and any elements connected in parallel with it.

Nodal Analysis With Voltage Source

In the figure shown, nodes 2 and 3 form a supernode.

Applying KCL on supernode, then we get

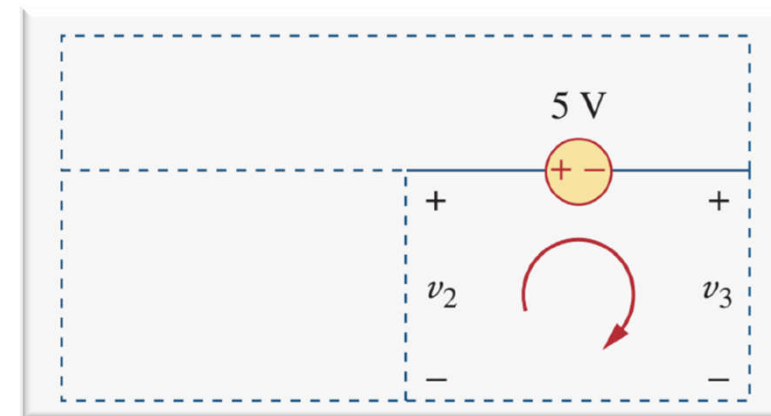
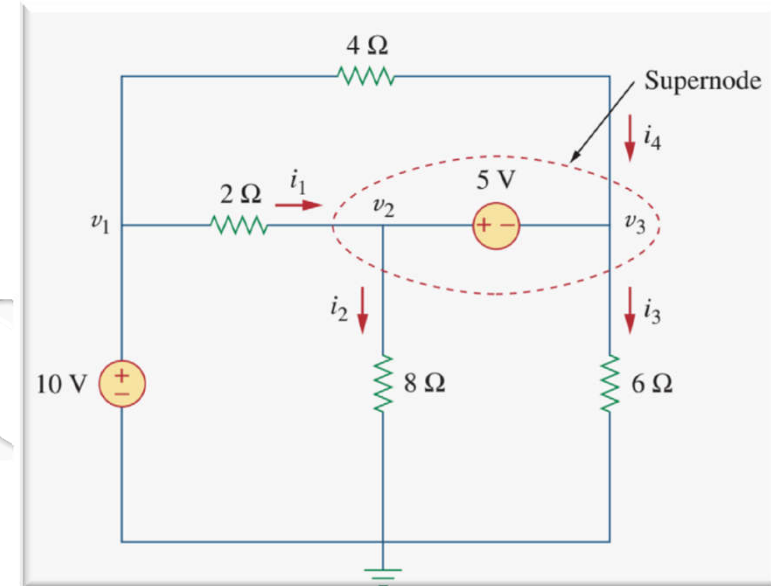
$$\frac{v_2}{8} + \frac{v_2 - v_1}{2} + \frac{v_3}{6} + \frac{v_3 - v_1}{4} = 0$$

Applying KVL on supernode, then we get

$$v_2 - v_3 = 5$$

Note the following properties of a supernode:

1. The voltage source inside the supernode provides a constraint equation needed to solve for the node voltages.
2. A supernode has no voltage of its own.
3. A supernode requires the application of both KCL and KVL.



Nodal Analysis With Voltage Source

Example: For the circuit shown, find the node voltages.

The supernode contains the 2V source, nodes 1 and 2, and the 10Ω resistor.

Applying KCL to the supernode gives

$$-2 + \frac{v_1}{2} + \frac{v_2}{4} + 7 = 0$$

$$-8 + 2v_1 + v_2 + 28 = 0$$

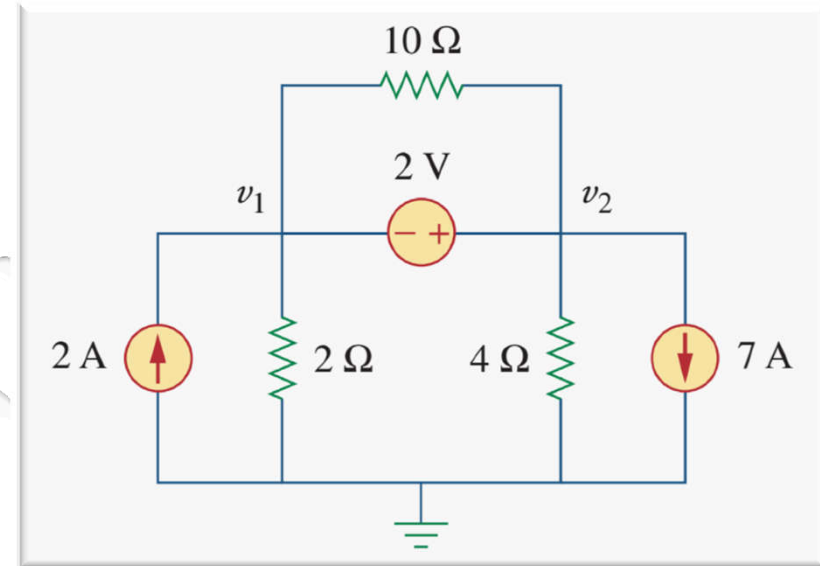
$$v_2 = -20 - 2v_1 \quad (1)$$

Applying KVL to the supernode gives

$$v_2 - v_1 = 2 \quad \text{or} \quad v_2 = v_1 + 2 \quad (2)$$

From eq.(1) and eq.(2)

$$v_2 = v_1 + 2 = -20 - 2v_1$$



$$3v_1 = -22$$

$$v_1 = -7.33\text{V}$$

$$v_2 = v_1 + 2 = -7.33 + 2 = -5.33\text{V}$$

Note that the 10- resistor does not make any difference because it is connected across the supernode.

Nodal Analysis With Voltage Source

Example: For the circuit shown, find the node voltages.

Node 1

$$8 + 3 + \frac{v_1 - v_2}{3} + \frac{v_1 - v_3}{4} = 0$$

$$0.5833v_1 - 0.3333v_2 - 0.25v_3 = -11 \quad (1)$$

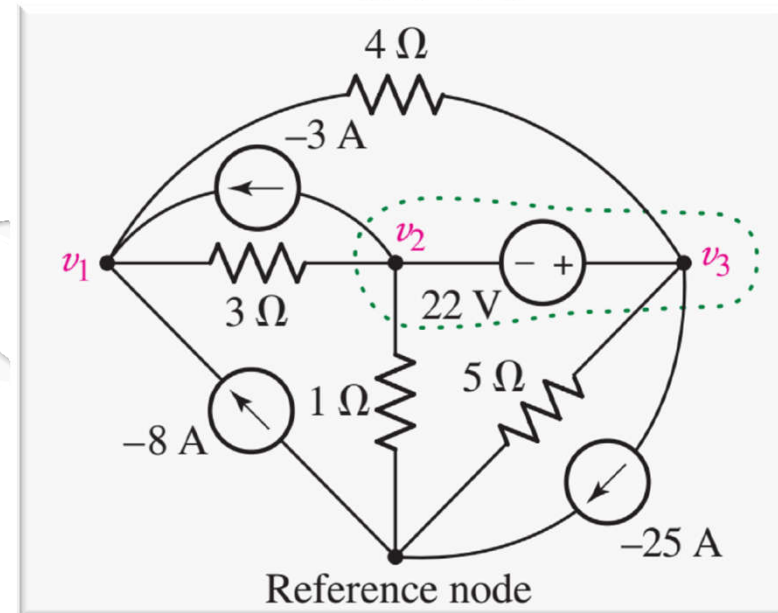
Supernode

$$-3 + \frac{v_2 - v_1}{3} + \frac{v_1}{1} - 25 + \frac{v_3 - v_1}{4} + \frac{v_3}{5} = 0$$

$$-0.5833v_1 + 1.3333v_2 + 0.45v_3 = 28 \quad (2)$$

$$v_3 - v_2 = 22 \quad (3)$$

Solve eq.(1), eq.(2) and eq.(3) to get v_1 , v_2 , and v_3 .



Nodal Analysis With Voltage Source

Example: For the circuit shown, find the node voltages.

Node 1

$$9.5v_1 - 2.5v_2 - 5v_3 = 5 \quad (1)$$

Node 2

$$v_2 = 18 \text{ V} \quad (2)$$

Supernode

$$v_3 - v_2 = 12 \quad (3)$$

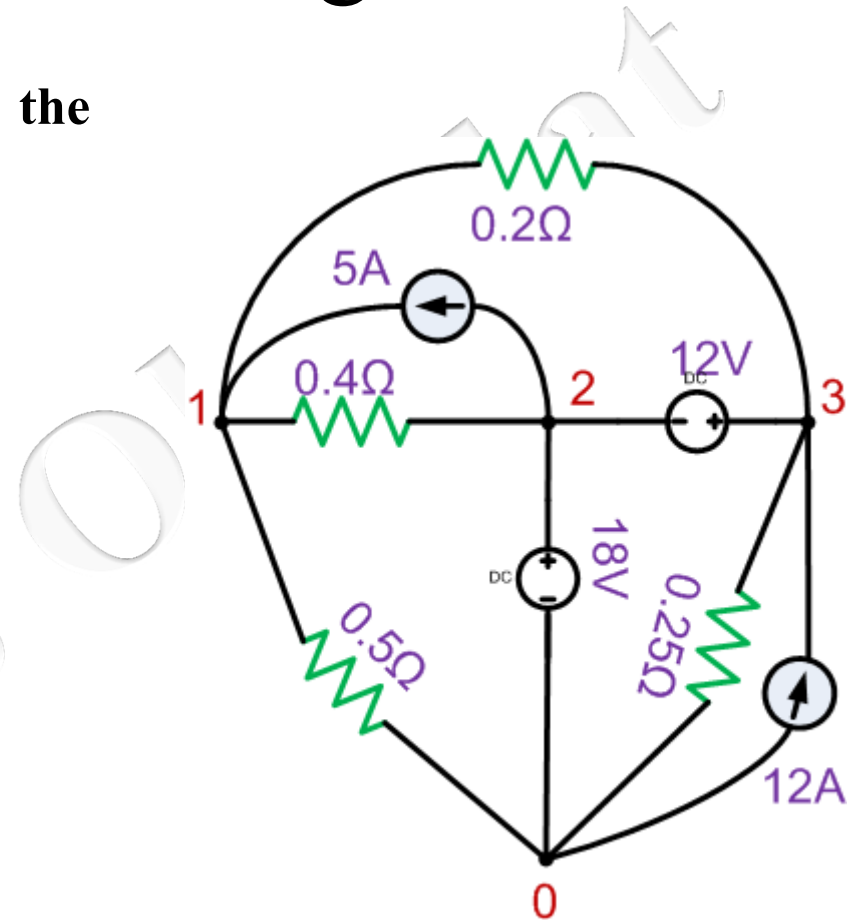
$$v_3 - 18 = 12$$

$$v_3 = 30 \text{ V}$$

Substitute v_2 and v_3 in eq.(1)

$$9.5v_1 - 2.5 \times 18 - 5 \times 30 = 5$$

$$v_1 = 21.5 \text{ V}$$



Nodal Analysis With Voltage Source

Example: For the circuit shown, find the node voltages.

Node 1

$$v_1 = -12V \quad (1)$$

Node 2

$$-14 + \frac{v_2 - v_1}{0.5} + \frac{v_2 - v_3}{2} = 0 \quad (2)$$

Supernode

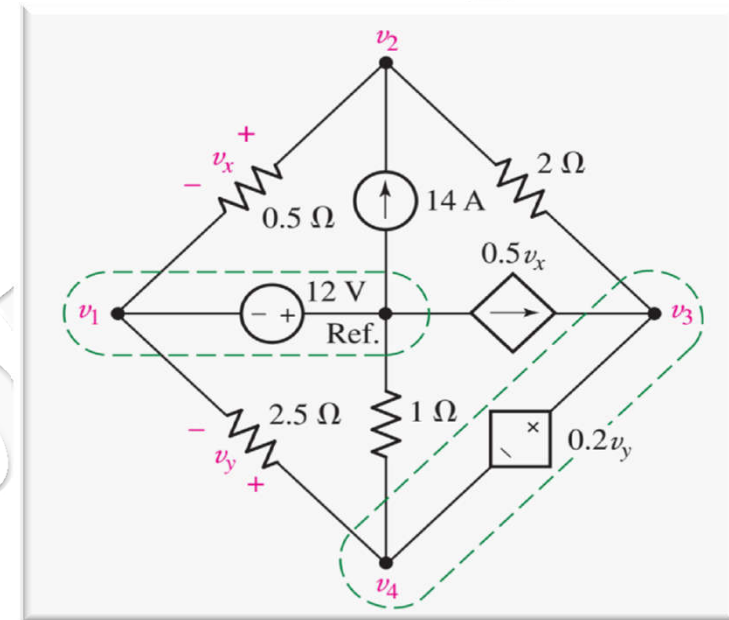
$$-0.5v_x + \frac{v_3 - v_2}{2} + \frac{v_4}{1} + \frac{v_4 - v_1}{2.5} = 0 \quad (3)$$

$$-0.5(v_2 - v_1) + \frac{v_3 - v_2}{2} + \frac{v_4}{1} + \frac{v_4 - v_1}{2.5} = 0 \quad (3)$$

$$v_3 - v_4 = 0.2v_y \quad (4)$$

$$v_3 - v_4 = 0.2(v_4 - v_1) \quad (4)$$

Rearrange these four equations then we have the following equations



$$v_1 = -12V \quad (5)$$

$$-2v_1 + 2.5v_2 - 0.5v_3 = 14 \quad (6)$$

$$0.1v_1 - v_2 + 0.5v_3 + 1.4v_4 = 0 \quad (7)$$

$$0.2v_1 + v_3 - 1.2v_4 = 0 \quad (8)$$

$$v_1 = -12V \quad v_2 = -4V$$

$$v_3 = 0V \quad v_4 = -42V$$

